TECHNOLOGY

Container Orchestration Using Kubernetes: Course-End Project

Deploy the Application Using the Kubernetes Dashboard

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Objectives

To deploy a multi-tier PHP and MySQL application using Kubernetes with specific configurations for user roles, storage, service verification, namespace restrictions, quota limits, and data management.

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Problem Statement and Motivation

Real-Time Scenario:

Karen is a DevOps engineer at a tech startup. Her team has developed a new application using PHP and MySQL. Now, it is her task to deploy that application.

The company plans to utilize Kubernetes for its robust container orchestration capabilities.

Karen must create a Kubernetes dashboard with specific configurations, user roles, storage, service verification, and data management.

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Industry Relevance

The following tools utilized in this project are widely applied in the industry:

1. **kubeadm:** A utility that offers **kubeadm init** and **kubeadm** join as efficient ways to bootstrap Kubernetes clusters. It focuses on bootstrapping rather than machine provisioning.

2. **kubectl:** A command-line interface for Kubernetes that allows execution of commands against Kubernetes clusters. It can be used for deploying applications, managing cluster resources, and viewing logs.

3. **kubelet:** An essential node agent present on every node in a Kubernetes cluster. It ensures that the containers described in the provided PodSpecs are running and healthy.

4. **Docker:** Docker is a tool designed to facilitate developers in building, sharing, and running applications in containers. It takes care of the setup, allowing developers to concentrate on the code.

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Tasks

The following tasks outline the application using Kubernetes:

1. Getting started with pods, services, and deployments
2. Creating and verifying the service
3. Creating a token and working on a dashboard
4. Configure the NFS-server for MySQL and WordPress deployment
5. Setting up the NFS client side
6. Creating and verifying the PV
7. Creating a secret for MySQL deployments secret data
8. Creating a configmap for WordPress deployment to store non-sensitive information

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Project Reference

- **Task 1:** Lesson 2
- **Task 2:** Lesson 6
- **Task 3:** Lesson 3
- **Task 4:** Lesson 4
- **Task 5:** Lesson 4
- **Task 6:** Lesson 7
- **Task 7:** Lesson 4
- **Task 8:** Lesson 4

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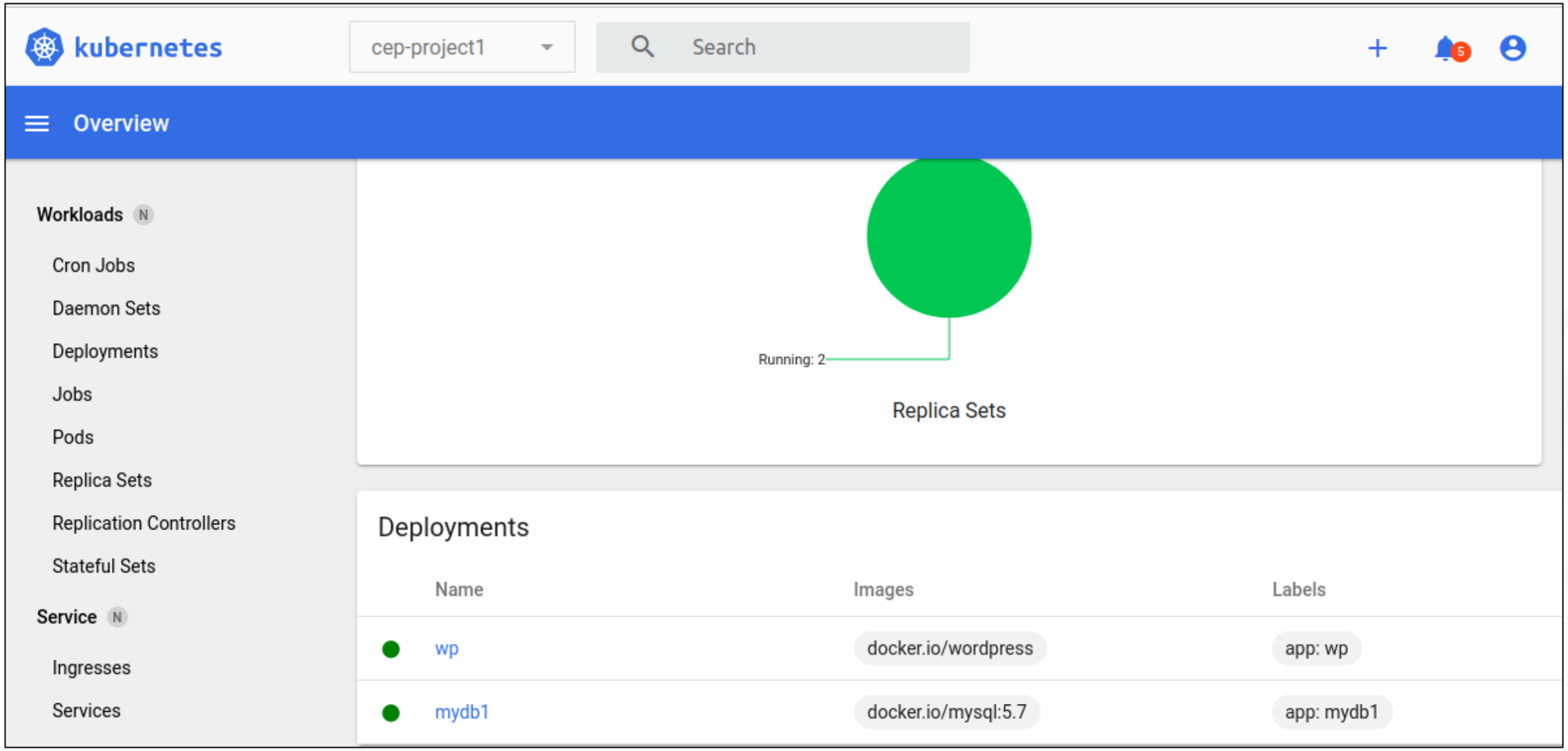
Sample Output Screenshots



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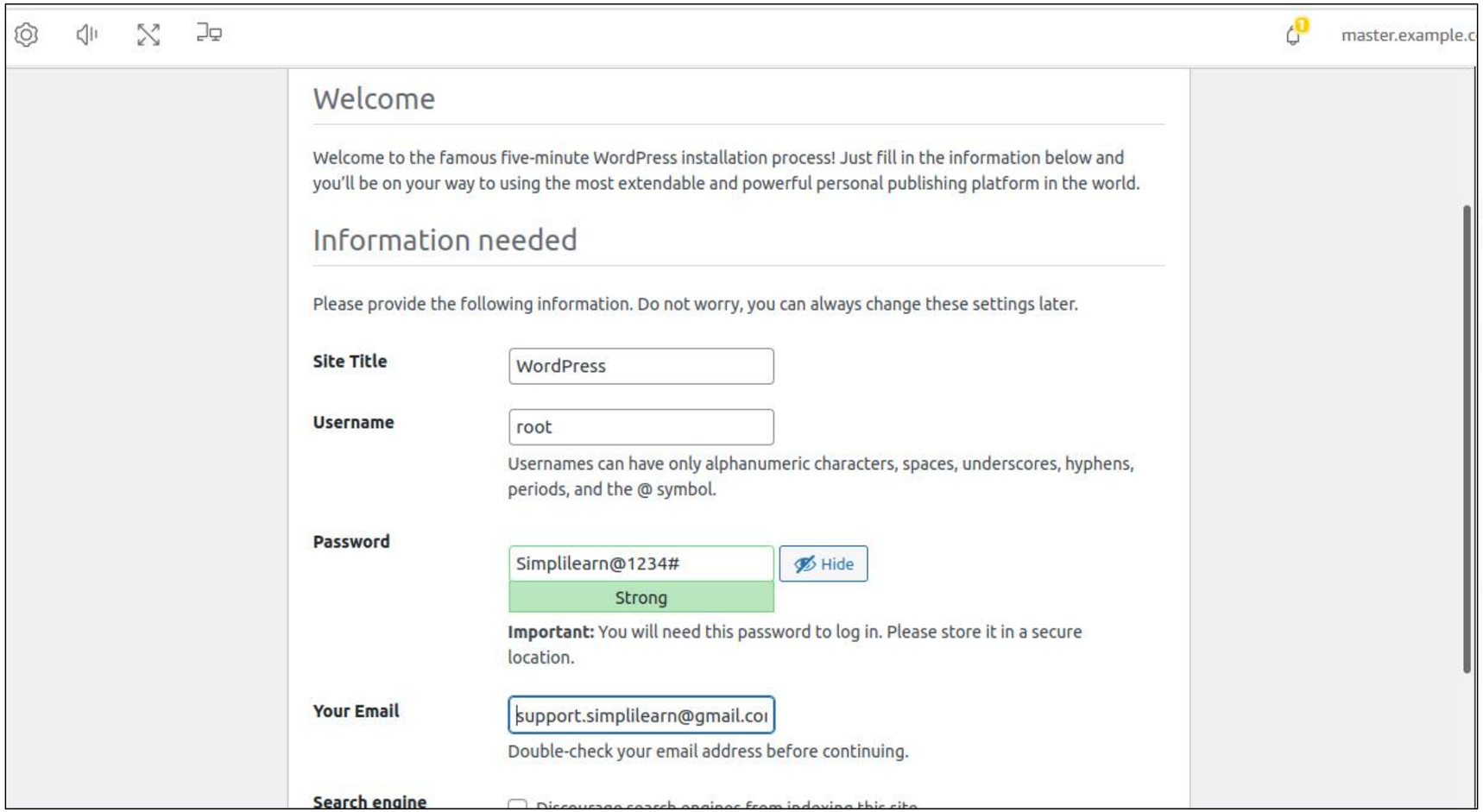

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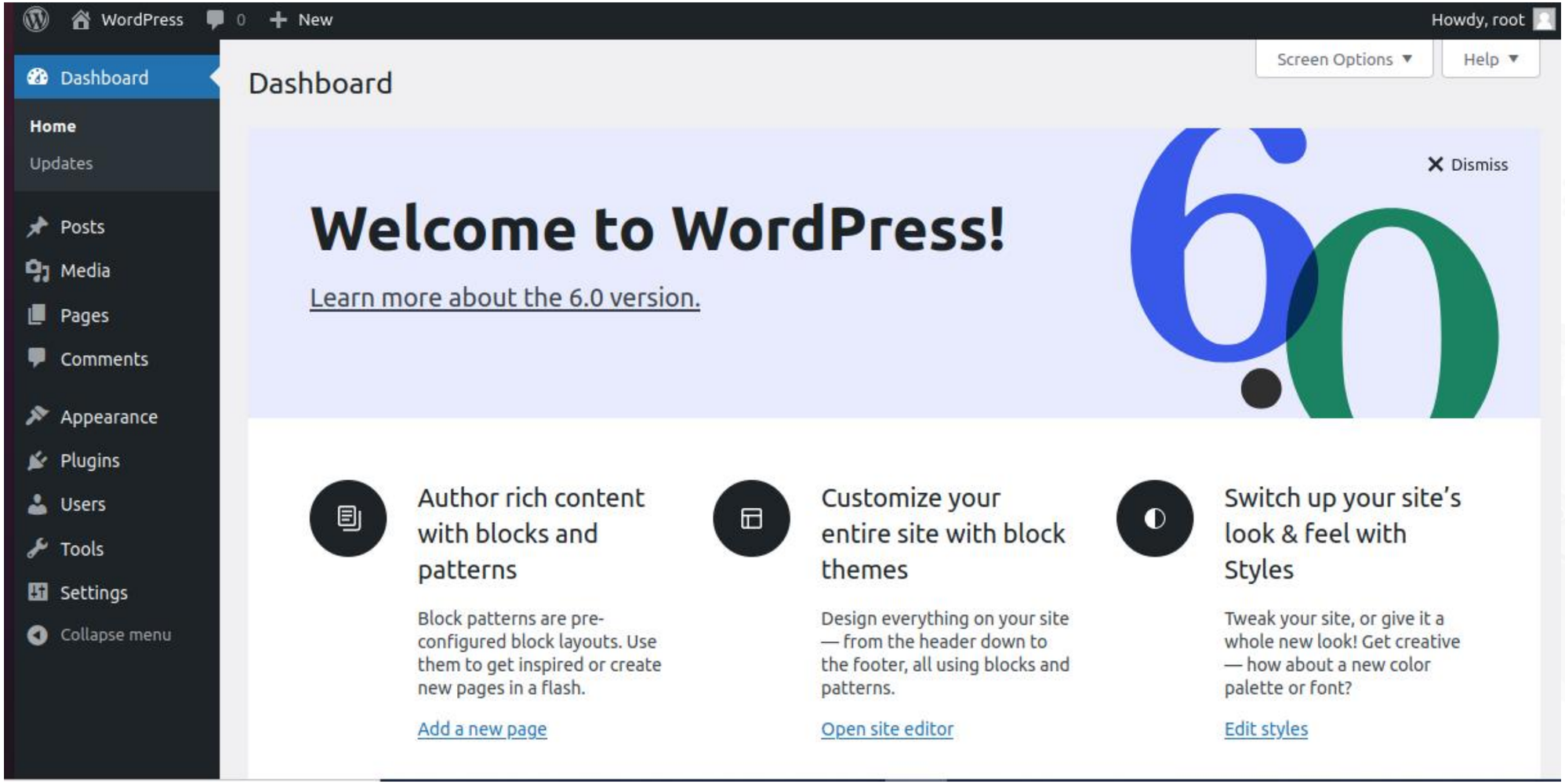

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Thank You

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